

Community assembly processes in a montane grassland are not driven by environmental factors

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Understanding what drives community composition is a central goal in ecology (HilleRisLambers et al., 2012). **Environmental filtering** may structure communities by excluding species which do not possess the traits to survive at a site. This assembly mechanism would create a pattern of trait convergence toward an optimum trait value (Gotzenberger et al., 2012). However, competition may also structure communities by competitively excluding species with similar traits. This would create a pattern of trait divergence, termed **limiting similarity** (Gotzenberger et al., 2012).

The role of environmental filtering in structuring communities is expected to increase with environmental severity, while limiting similarity is expected to be the dominant assembly mechanism in productive environments (Louthan, Doak & Angert, 2015). However, considerable variation in assembly mechanisms exist at fine scales within communities (Stark et al., 2017). Here we ask what drives this variation?

Questions:

- 1. Does the community assembly mechanism change with elevation?
- 2. What drives fine-scale variation in community assembly?

Methods

Species composition, functional trait (plant height and specific leaf area [SLA]) and environmental data were collected from three sites in the Maloti-Drakensberg spanning an elevational gradient of 1000 m (Figure 1). Data was collected in sampling grids of 8 x 20 m that were divided into 160 contiguous 1 x 1 m cells. To determine whether the trait diversity (estimated with RaoQ) of a cell was different from a null model of random assembly, we calculated a standardized effect size (SES). **SES > 0** indicates that trait diversity was higher than expected from a null model (indicative of **limiting similarity**), while **SES < 0** indicates that trait diversity was lower than expected (indicative of **environmental filtering**). We used generalized linear mixed effect models to test whether elevation or microenvironmental variables (rock cover, aspect, soil moisture, soil depth and slope height) were related to the SES. Grid was included as a random effect to account for spatial structure in the dataset.

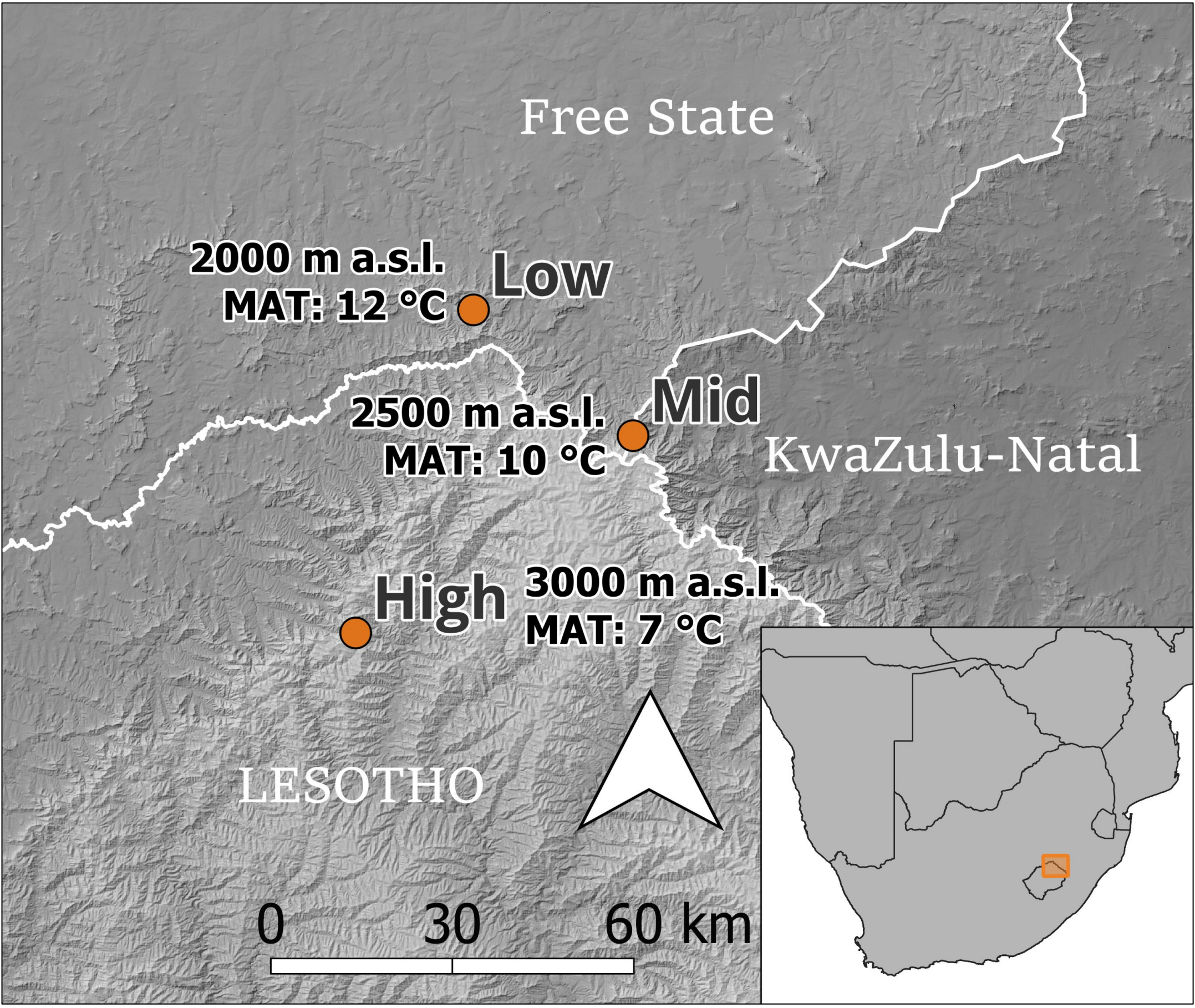


Figure 1: Sites along an elevational gradient in the Maloti-Drakensberg of South Africa and Lesotho.

Results and discussion

Models explained little variation in the SES of plant height (marginal $R^2 = 0.03$, conditional $R^2 = 0.07$) and SLA (marginal $R^2 = 0.15$, conditional $R^2 = 0.27$). Elevation was the most important variable in predicting the deviation of diversity in height and SLA from random assembly (Figure 2). Environmental filtering was the dominant assembly mechanism for plant height

and SLA at all elevations (Figure 3). Environmental filtering for plant height was strongest at the mid elevation, selecting for plants of intermediate height (Figure 4). Environmental filtering for SLA was strongest at the low and high elevation, selecting for low SLA (Figure 4).

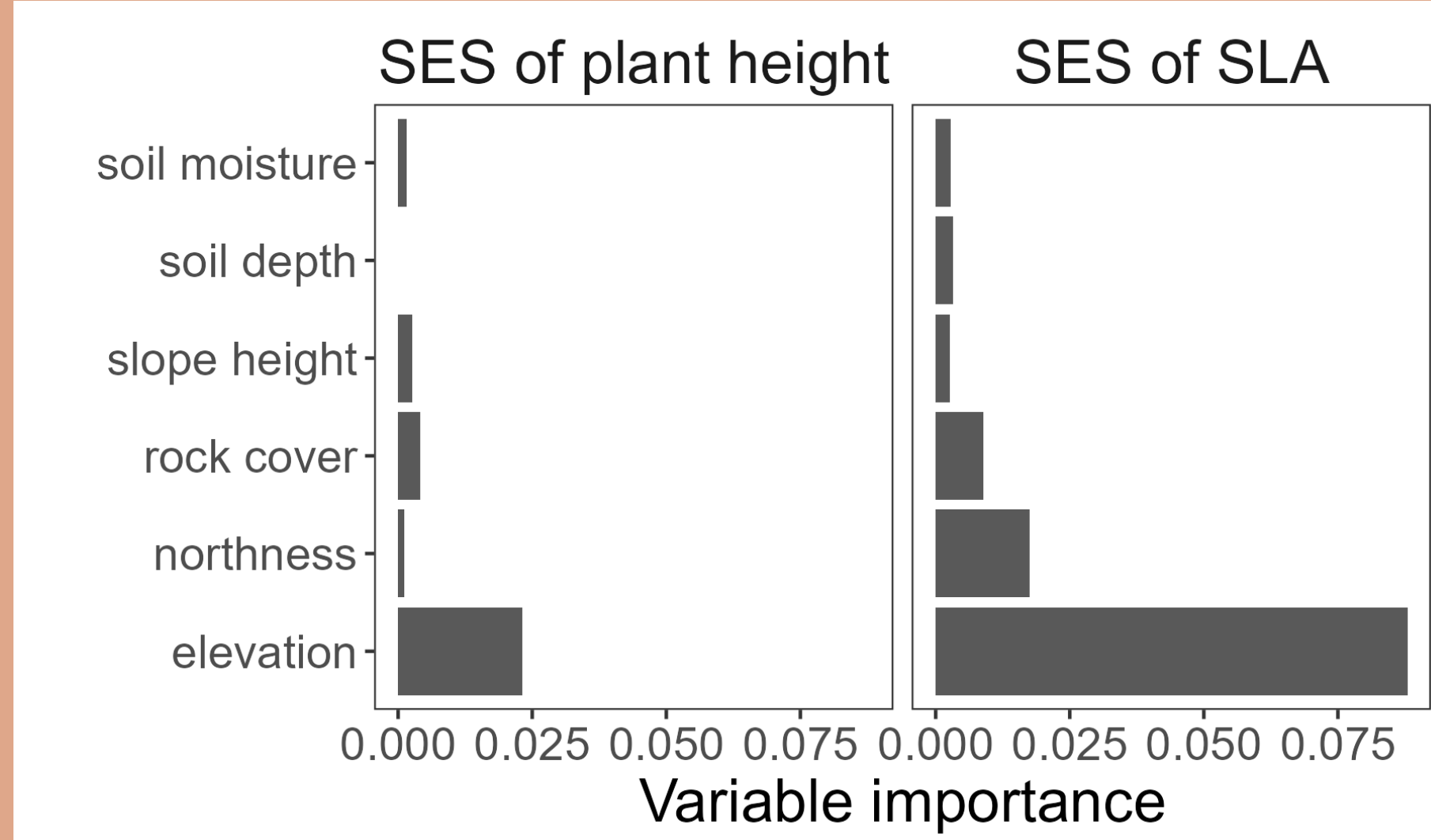


Figure 2: Importance of environmental variables in predicting the deviation of diversity in height (SES of plant height) and diversity in SLA (SES of SLA) from random assembly.

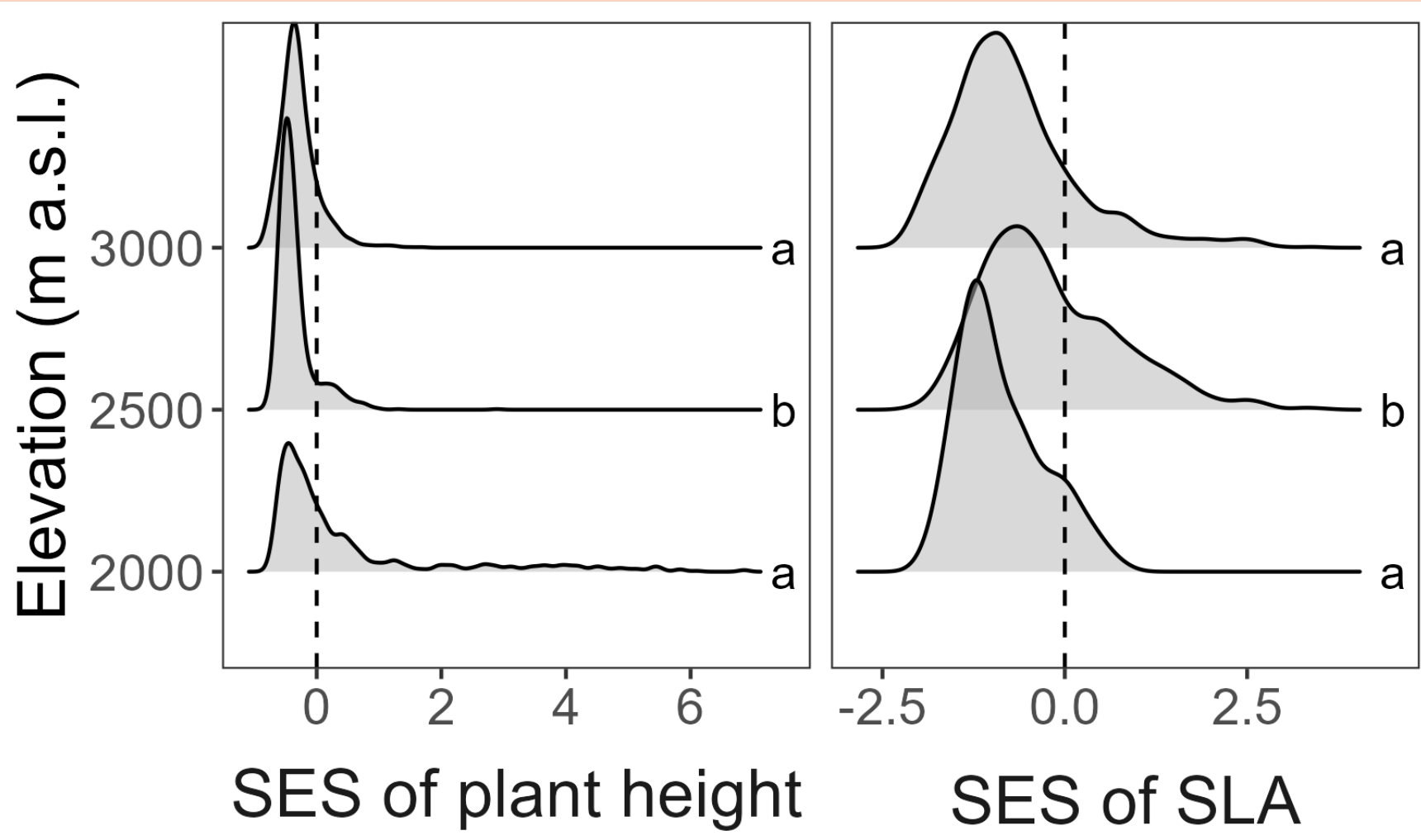


Figure 3: Distribution of SES calculated from plant height and SLA at low, mid and high elevation. Letters reflect significant differences in the mean SES of each elevation.

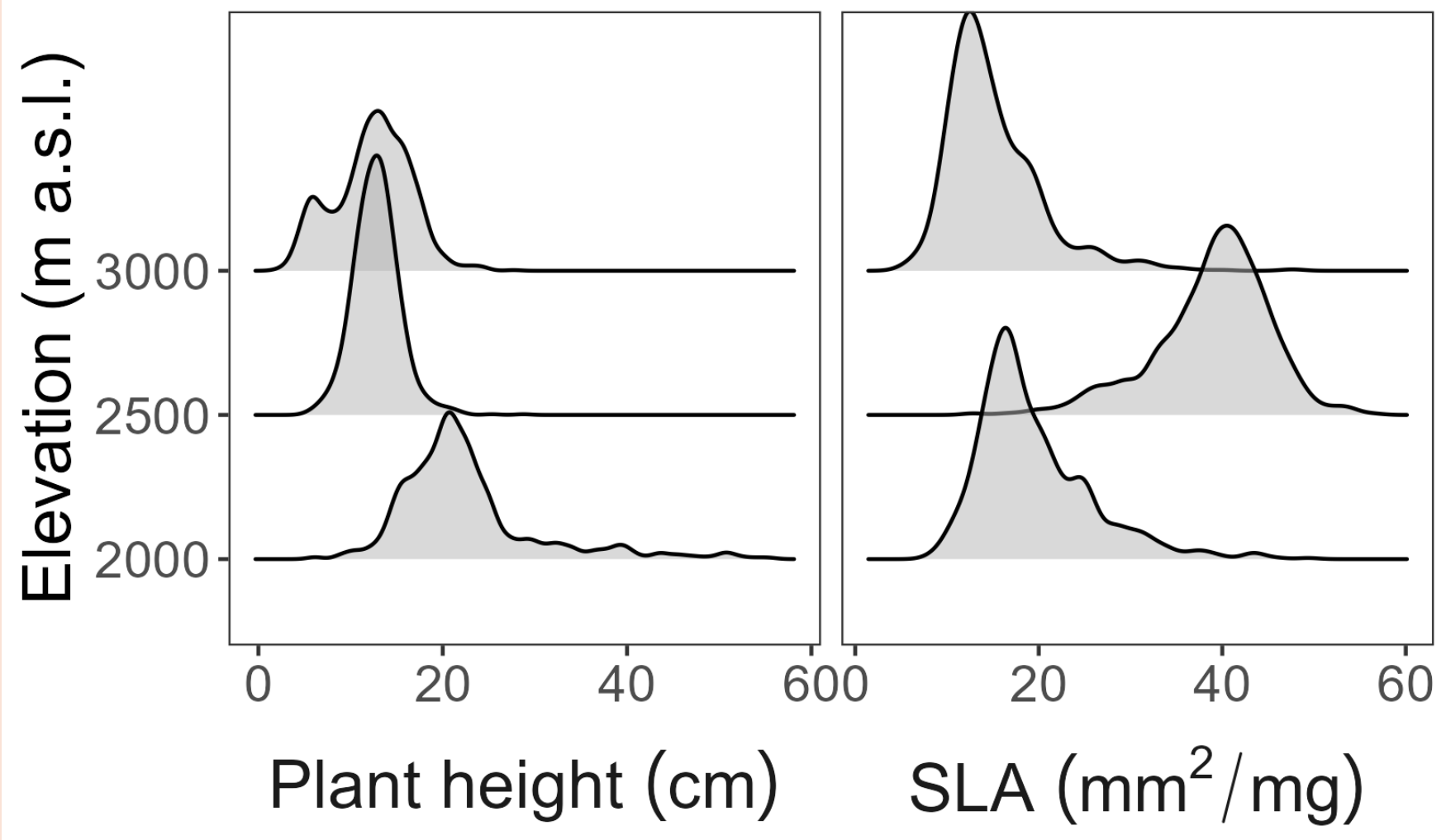


Figure 4: Community weighted mean traits of cells at low, mid and high elevation

Environmental filtering was the dominant assembly mechanism for all traits at all elevations. Environmental filtering shifted communities toward increasingly shorter height with elevation, likely due to the increased cold tolerance of short-statured plants. The SLA of communities were filtered toward conservative leaf-economic strategies at the low and high elevation, but selection shifted to a more acquisitive leaf-economic strategy at the mid elevation.

This may reflect a relaxation of temperature limitations on growth, or a competitive advantage of high SLA at the mid elevation. Microenvironmental variation only weakly explained small-scale variation in SES, suggesting that small scale variation in community assembly signatures may be the result of stochastic processes.

References:

Gotzenberger et al., (2012). *Biological Reviews*, 87(1), 111-127. doi:10.1111/j.1469-185X.2011.00187.x
HilleRisLambers et al., (2012). In D. J. Futuyma (Ed.), *Annual Review of Ecology, Evolution, and Systematics*, Vol 43 (Vol. 43, pp. 227-248).
Louthan et al., (2015). *Trends in Ecology & Evolution*, 30(12), 780-792. doi:10.1016/j.tree.2015.09.011
Stark et al., (2017). *Oikos*, 126(11), 1650-1659. doi:10.1111/oik.04311

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